

BIRNI NKONNI, Niger

WMO: 610750

Lat: 13.800N

Lon: 5.250E

Elev: 896

StdP: 14.23

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 60.6	(c) 62.7	(d) 18.6	(e) 14.5	(f) 80.8	(g) 21.5	(h) 16.7	(i) 80.8	(j) 18.1	(k) 76.4	(l) 15.8	(m) 75.7	(n) 2.4	(o) 100	(p) 0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	25.5	108.6	70.7	107.1	70.8	105.5	70.8	79.9	94.4	79.2	93.5	78.7	92.9	4.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	143.2	84.9	76.0	140.3	84.6	75.4	137.4	84.3	44.2	94.3	43.5	93.8	42.9	93.2	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.6	11.2	9.4	DB	55.8	112.0	1.8	0.9	54.4	112.7	53.3	113.2	52.3	113.8	51.0	114.4
			WB	42.5	81.5	1.4	1.6	41.5	82.6	40.6	83.6	39.8	84.5	38.8	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.7	76.6	82.5	89.0	93.8	94.1	89.9	85.4	82.8	85.0	87.6	83.5	77.8
	DBStd	6.40	4.12	4.79	3.87	2.78	3.39	3.95	3.55	2.91	3.28	2.40	3.13	3.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13017	823	910	1209	1315	1369	1196	1097	1016	1049	1165	1006	862
	CDD65	7542	358	490	744	865	904	746	632	551	599	700	556	397
	CDH74	104014	3709	6238	11307	14240	14681	11159	8225	6240	7432	9545	7072	4166
	CDH80	62722	1767	3601	7408	10014	10264	7017	4268	2698	3787	5653	4203	2042
Wind	WSAvg	3.4	3.9	4.0	3.8	3.4	4.0	4.5	4.0	2.9	2.4	2.1	2.4	3.7
Precipitation	PrecAvg	19.0	0.0	0.0	0.0	0.2	1.1	2.5	5.1	6.5	3.2	0.5	0.0	0.0
	PrecMax	25.3	0.2	2.0	0.5	1.5	4.4	6.2	11.2	13.6	12.6	4.2	0.0	0.0
	PrecMin	11.4	0.0	0.0	0.0	0.0	0.0	0.3	1.5	0.0	0.6	0.0	0.0	0.0
	PrecStd	3.8	0.0	0.3	0.1	0.3	1.0	1.4	2.1	2.5	2.0	0.8	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.3	105.6	108.3	110.8	110.1	105.6	100.4	96.2	100.6	103.9	102.6	98.9
		MCWB	63.8	66.3	67.3	69.8	73.3	76.5	77.4	78.7	77.9	70.1	67.4	65.3
	2%	DB	95.6	102.3	106.7	109.2	108.1	103.3	97.9	94.0	97.9	102.1	101.1	96.0
		MCWB	62.4	65.6	66.6	69.6	74.2	76.4	77.1	78.0	78.2	70.9	66.8	63.6
	5%	DB	92.7	99.7	105.0	107.7	106.4	101.3	95.9	92.4	95.8	100.4	99.3	93.4
		MCWB	61.0	64.6	66.2	70.1	74.5	76.6	76.9	77.6	78.1	71.7	66.0	62.1
	10%	DB	89.7	96.4	102.9	106.0	104.5	99.4	93.9	90.5	93.7	98.6	97.1	90.5
		MCWB	59.8	63.5	65.6	70.0	74.9	76.4	76.7	77.2	77.9	72.5	64.9	60.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	71.5	73.7	78.1	80.2	80.7	79.9	80.0	80.9	79.9	72.4	66.5
		MCDB	94.4	94.4	98.4	101.8	101.0	97.7	93.1	92.1	93.4	94.8	95.6	95.7
	2%	WB	63.9	68.3	71.4	76.7	79.0	79.2	78.8	79.1	79.9	78.9	69.1	64.9
		MCDB	92.8	96.3	97.9	99.8	99.3	96.0	92.8	90.5	92.5	93.2	94.8	93.4
	5%	WB	61.9	66.0	68.4	75.7	78.1	78.4	78.1	78.4	79.2	78.1	67.4	63.3
		MCDB	90.6	96.4	100.8	98.4	98.4	94.9	91.4	89.0	91.6	92.0	94.7	91.0
	10%	WB	60.2	64.2	66.5	74.6	77.3	77.7	77.5	77.8	78.6	77.3	66.2	61.6
		MCDB	88.4	95.2	100.6	97.1	97.5	93.9	90.1	87.9	90.7	90.9	94.0	88.6
Mean Daily Temperature Range	5% DB	MDBR	26.5	26.9	27.8	25.5	21.2	18.5	16.1	14.7	17.9	23.8	28.1	26.2
		MCDBR	30.3	30.4	30.4	27.8	23.7	20.2	18.4	17.2	20.4	26.5	30.1	29.4
		MCWBR	12.8	12.0	11.4	10.6	7.5	5.1	4.7	5.0	5.9	7.6	10.3	12.1
	5% WB	MCDBR	29.8	29.0	28.8	22.9	20.4	18.9	16.8	16.1	18.9	21.0	29.9	29.3
		MCWBR	13.1	12.4	12.5	9.1	6.6	5.7	5.3	5.6	6.6	6.4	10.8	12.4
Clear-Sky Solar Irradiance	taub		0.579	0.636	0.704	0.735	0.725	0.699	0.634	0.603	0.593	0.621	0.544	0.539
	taud		1.720	1.570	1.456	1.469	1.509	1.582	1.760	1.817	1.823	1.700	1.759	1.779
	Ebn at Noon		229	223	212	206	204	207	222	232	235	224	236	235
	Edh at Noon		73	88	100	99	93	86	72	69	69	76	69	67
All-Sky Solar Radiation	RadAvg		1740	1903	2004	2062	2007	1919	1870	1793	1950	1926	1821	1704
	RadStd		75	78	109	78	63	80	62	71	62	55	46	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page